

Latency measurement implementations - results comparison

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I have finished implementing 'replay' functionality. It is a feature that allows to literally replay the flow but using the input data stored in a file rather than live accounts updated from a blockchain.

Here is a screenshot with a calculated mean value for duration values:

Note that 'usdt_to_other_amount_out_calculation', 'other_to_usdt_amount_out_calculation' and 'profit_calculation' are stubbed, i.e the values are the same for both on-chain and replay logic - these values are not assessed.

I focused on 'account_mutation_dispatch_and_send_to_flow', 'usdt_to_other_warm_up' and 'other_to_usdt_warm_up' values and you can see how I achieved some performance improvement.

MEAN	on-chain	replay	diff %	faster?
account_mutation_arrival	4683334	351198	13.335599296197682	true
account_mutation_dispatch_and_send_to_flow	24778	2195	11.290593594642477	true
dispatch_immediately				(reading from file - not significant change, just let it be)
usdt_to_other_warm_up	2031465	893292	2.2741334561808503	true
usdt_to_other_amount_out_calculation	9807204959	9802223972	1.0005533862575398	true
other_to_usdt_warm_up	261756	2788	93.89291639964131	true
other_to_usdt_amount_out_calculation	980752243	9802839404	1.000213176645766	true
profit_calculation	1806843831	1801500875	1.0045353499721337	true
total_reaction	19826789141	19803754585	1.001212105510416	true

So, the table above compares two different logics, two different implementations of the same features.

Next, I transitioned 'on-chain' logic onto new 'rails', meaning that on-chain logic and replay logic use the same code, but again, on-chain logic listens on-chain data updates, but replay logic replays the stored account changes (stored during on-chain logic ran). **There are visible improvements in duration values for on-chain logic, but again, the values are not the same and this part of my investigation is the most demanding!**

Here is a screenshot of logs when both logics use the same code:

MEAN	on-chain	replay	diff %	abs diff, ns	faster?
account_mutation_arrival	4828349	448258	8.986680201351344	3580091	true
account_mutation_dispatch_and_send_to_flow	3161	3084	1.024997042418294	77	true
usdt_to_other_warm_up	363757	1522078	0.2389867962394889	1158321	false
usdt_to_other_amount_out_calculation	9084414958	9016506645	0.9986589388015107	12801687	false
other_to_usdt_warm_up	13379	3476	3.8488321463004986	9903	true
other_to_usdt_amount_out_calculation	9084414958	9089590368	0.9995365078690285	4175411	false
profit_calculation	1003206195	1001425505	1.0017781551051208	1780690	true
total_reaction	19015349516	19025211798	0.9994816203633823	9862282	false

All values are in nanoseconds and I added the absolute difference (it is in nanoseconds as well).

For now I can think that the **difference exists because of operational environment, not because of the logic itself**. This brings me closer to **High-Frequency Trading (HFT) systems latency studying**. I've just scratched the surface of this field, but some conclusions may be made for now: the difference between on-chain and replay logics was caused by either one or all factors, and other factors that are not taken into account yet:

1. Incorrect cache memory usage - both logics use different levels of cache memory. Along with this, CPU may attempt to access a remote memory belonging to another CPU, which might be 1,5-3x slower.
2. I am using Tokio which is not used in professional HFT systems because of Tokio runtime which orchestrates threads as it thinks is ok, so the multiple threads should be managed in other way.

Also, the context switching should not be a case when using Tokio, because it uses 'green' threads and async approach, and 'await' points are the same in both logics.

There is a book **'The Rust Performance Book'** and I hope to find some techniques to apply in it. Also, I would like to move closer to HFT systems style of latency measurement and system architecture.

I analyzed mean, p50, p95, p99, min, max values as well. The results are below - **the first group of tables** represents the results when both logics use different implementations, and the **second group of tables** is about the same implementations for both logics.

First group of tables:

'Replay' and 'On-chain' logic use different logic - 'Replay' logic is considered as improved

MEAN	on-chain	replay	diff %	faster?
account_mutation_arrival	4683334	369733	12.66680840272299	true
account_mutation_dispatch_and_send_to_flow	24778	2279	10.87082494175753	true
usdt_to_other_warm_up	2831465	1231433	1.64967580308194	true
usdt_to_other_amount_out_calculation	9007204959	9002076803	1.000569663593064	true
other_to_usdt_warm_up	261756	2871	91.16294318854685	true
other_to_usdt_amount_out_calculation	9007532743	9002357762	1.000574847380472	true
profit_calculation	1006043031	1001303535	1.004733326569052	true
total_reaction	19026789141	19003234144	1.0012395257050566	true
p50	on-chain	replay	diff %	faster?
account_mutation_arrival	254719	352767	0.7220601700272418	false
account_mutation_dispatch_and_send_to_flow	24447	2253	10.8508655126498	true
usdt_to_other_warm_up	1175551	308735	3.807637618021928	true
usdt_to_other_amount_out_calculation	9009364991	9009364991	1	false
other_to_usdt_warm_up	258815	2531	102.2580007902015	true
other_to_usdt_amount_out_calculation	9009364991	9009364991	1	false
profit_calculation	1002962943	1001390079	1.0015706806298408	true
total_reaction	19025362943	19008585727	1.0008826125331445	true
p95	on-chain	replay	diff %	faster?
account_mutation_arrival	19300351	592895	32.55273024734565	true
account_mutation_dispatch_and_send_to_flow	58719	3171	18.517502365184484	true
usdt_to_other_warm_up	5148671	585215	8.797913587314063	true
usdt_to_other_amount_out_calculation	10007609343	10007609343	1	false
other_to_usdt_warm_up	414875	4571	90.78420227740181	true
other_to_usdt_amount_out_calculation	10015997951	10007609343	1.000838222967393	true
profit_calculation	1028653055	1002438655	1.0261506276411498	true
total_reaction	21055406079	21021851647	1.0015961691940105	true
p99	on-chain	replay	diff %	faster?
account_mutation_arrival	34570239	712703	48.50581378217855	true
account_mutation_dispatch_and_send_to_flow	73855	3791	19.481667106304407	true
usdt_to_other_warm_up	9773855	769535	12.699948670300897	true
usdt_to_other_amount_out_calculation	10032775167	10007609343	1.002514668902179	true
other_to_usdt_warm_up	513791	10647	48.256879872264484	true
other_to_usdt_amount_out_calculation	10041163775	10007609343	1.0033528918695722	true
profit_calculation	1053294591	1004011519	1.0490861619287857	true
total_reaction	21088960511	21021851647	1.003192338388021	true
MAX	on-chain	replay	diff %	faster?
account_mutation_arrival	245891071	1653759	148.686157414714	true
account_mutation_dispatch_and_send_to_flow	1498111	9047	165.59201945396265	true
usdt_to_other_warm_up	20479999	499384319	0.0410104967673204	false
usdt_to_other_amount_out_calculation	10099884031	10007609343	1.0092204526413238	true
other_to_usdt_warm_up	749055	125311	5.977567811285522	true
other_to_usdt_amount_out_calculation	10099884031	10007609343	1.0092204526413238	true
profit_calculation	1102053375	1006108671	1.09536216789051	true
total_reaction	21307064319	21508390911	0.9906396255845882	false
MIN	on-chain	replay	diff %	faster?
account_mutation_arrival	99136	125440	0.7903061224489796	false
account_mutation_dispatch_and_send_to_flow	2354	768	3.0651041666666665	true
usdt_to_other_warm_up	69312	114688	0.6043526785714286	false
usdt_to_other_amount_out_calculation	7998537728	7998537728	1	false
other_to_usdt_warm_up	64736	512	126.4375	true
other_to_usdt_amount_out_calculation	7998537728	7998537728	1	false
profit_calculation	1000341504	999817216	1.0005243838489775	true
total_reaction	17003708416	16995319008	1.0004935834155972	true

Second group of tables:
'Replay' and 'On-chain' logics use the same code

MEAN	on-chain	replay	diff %	abs diff, ns	faster?
account_mutation_arrival	4828349	448258	8.986680201351344	3580091	true
account_mutation_dispatch_and_send_to_flow	3161	3084	1.024997042418294	77	true
usdt_to_other_warm_up	363757	1522078	0.2389867962394889	1158321	false
usdt_to_other_amount_out_calculation	9004414958	9016506645	0.9986589388015107	12091687	false
other_to_usdt_warm_up	13379	3476	3.8488321463004986	9903	true
other_to_usdt_amount_out_calculation	9004414958	9008598368	0.9995365078699285	4175411	false
profit_calculation	1003206195	1001425505	1.0017781551051208	1780690	true
total_reaction	19015349516	19025211798	0.9994816203633823	9862282	false
p50	on-chain	replay	diff %	abs diff, ns	faster?
account_mutation_arrival	3600383	370943	9.706027610711079	3229440	true
account_mutation_dispatch_and_send_to_flow	2443	2997	0.8151484818151484	554	false
usdt_to_other_warm_up	355327	1430527	0.2483888804615362	1075200	false
usdt_to_other_amount_out_calculation	9009364991	9009364991	1	0	false
other_to_usdt_warm_up	12583	3039	4.140506745640013	9544	true
other_to_usdt_amount_out_calculation	9009364991	9009364991	1	0	false
profit_calculation	1002438655	1001914367	1.0005232062380942	524288	true
total_reaction	19025362943	19088585727	1.0008826125331445	16777216	true
p95	on-chain	replay	diff %	abs diff, ns	faster?
account_mutation_arrival	12607487	871423	14.467700531200117	11736064	true
account_mutation_dispatch_and_send_to_flow	6135	4255	1.4418331374853115	1880	true
usdt_to_other_warm_up	542719	3882239	0.17607946690701143	2539520	false
usdt_to_other_amount_out_calculation	10007609343	10007609343	1	0	false
other_to_usdt_warm_up	17471	4235	4.125383707201889	13236	true
other_to_usdt_amount_out_calculation	10007609343	10007609343	1	0	false
profit_calculation	1008205823	1002438655	1.005753138081053	5767168	true
total_reaction	21021851647	21021851647	1	0	false
p99	on-chain	replay	diff %	abs diff, ns	faster?
account_mutation_arrival	19841023	978943	20.267802109009413	18862080	true
account_mutation_dispatch_and_send_to_flow	12975	4679	2.7730284248771104	8296	true
usdt_to_other_warm_up	708415	3481599	0.20117624114666852	2781184	false
usdt_to_other_amount_out_calculation	10007609343	10007609343	1	0	false
other_to_usdt_warm_up	34719	10631	3.26582635688082	24088	true
other_to_usdt_amount_out_calculation	10015997951	10007609343	1.000838222967393	8388608	true
profit_calculation	1022885887	1002438655	1.020397489560097	20447232	true
total_reaction	21038628863	21021851647	1.0007980845970053	16777216	true
MAX	on-chain	replay	diff %	abs diff, ns	faster?
account_mutation_arrival	19841023	978943	20.267802109009413	18862080	true
account_mutation_dispatch_and_send_to_flow	12975	4679	2.7730284248771104	8296	true
usdt_to_other_warm_up	730111	3674111	0.1987177306292597	2944000	false
usdt_to_other_amount_out_calculation	10032775167	10007609343	1.002514668902179	25165824	true
other_to_usdt_warm_up	60575	73023	0.8295331607849582	12448	false
other_to_usdt_amount_out_calculation	10041163775	10007609343	1.0033528918695722	33554432	true
profit_calculation	1029177343	1002438655	1.0266736401939727	26738688	true
total_reaction	21072183295	21021851647	1.0023942537910158	50331648	true
MIN	on-chain	replay	diff %	abs diff, ns	faster?
account_mutation_arrival	25168	228736	0.11003077783995523	203568	false
account_mutation_dispatch_and_send_to_flow	1594	1552	1.027061855670103	42	true
usdt_to_other_warm_up	115328	222080	0.5193003573487032	106752	false
usdt_to_other_amount_out_calculation	7998537728	8015314944	0.9979068550497122	16777216	false
other_to_usdt_warm_up	6024	488	12.344262295081966	5536	true
other_to_usdt_amount_out_calculation	7998537728	8015314944	0.9979068550497122	16777216	false
profit_calculation	999817216	999817216	1	0	false
total_reaction	17003708416	17037262848	0.9980305268340719	33554432	false